

## Case name: 500 N

### Description:

Tue Mar 31 11:47:42 2026

**Table 1. Propellant Specification**

| Component                           | Temperature, K | Pressure, MPa | Enthalpy, kJ/mol | Enthalpy, kJ/kg | Mass fraction |
|-------------------------------------|----------------|---------------|------------------|-----------------|---------------|
| C <sub>2</sub> H <sub>5</sub> OH(L) | 298.1          | 0.1013        | -277.5087        | -6023.834       | 0.1250000     |
| H <sub>2</sub> O(L)                 | 298.1          | 0.1013        | -285.8288        | -15865.907      | 0.0416667     |
| H <sub>2</sub> O <sub>2</sub> (L)   | 298.1          | 0.1013        | -187.7791        | -5520.532       | 0.8333333     |
| Total                               |                |               | -203.7058        | -6014.491       |               |

Propellant exploded formula:

C<sub>0.183798</sub> H<sub>2.367596</sub> O<sub>1.829767</sub> (based on 1 mole)  
C<sub>4.195185</sub> H<sub>54.040378</sub> O<sub>41.764437</sub> (by mass %)  
 $\alpha$ : 1.5048534 (oxidizer excess coefficient)  
O/F: 5.0000000  
O/F<sub>0</sub>: 3.3225827 (stoichiometric)  
 $\rho$ : kg/m<sup>3</sup>

**Table 2. Combustion Properties**

| Parameter                          | Injector     | Nozzle inlet | Nozzle throat | Nozzle exit Unit                  |
|------------------------------------|--------------|--------------|---------------|-----------------------------------|
| Pressure                           | 2.0000       | 1.9899       | 1.1402        | 0.0887 MPa                        |
| Temperature                        | 2464.9251    | 2464.0785    | 2290.6646     | 1540.6929 K                       |
| Enthalpy                           | -134224.6520 | -134278.2823 | -145712.8224  | -186821.0236 J/mol                |
|                                    | -6014.4916   | -6016.8251   | -6509.2131    | -8312.0480 kJ/kg                  |
| Entropy                            | 262.3562     | 262.3802     | 263.1855      | 264.2472 J/(mol·K)                |
|                                    | 11.7560      | 11.7569      | 11.7569       | 11.7569 kJ/(kg·K)                 |
| Internal energy                    | -154719.2026 | -154765.7939 | -164758.4892  | -199631.0713 J/mol                |
|                                    | -6932.8348   | -6934.8423   | -7360.0119    | -8881.9931 kJ/kg                  |
| Specific heat (p=const)            | 3.1583       | 3.1574       | 2.8578        | 2.1573 kJ/(kg·K)                  |
| Specific heat (V=const)            | 2.7305       | 2.7296       | 2.4537        | 1.7866 kJ/(kg·K)                  |
| Gamma                              | 1.1567       | 1.1567       | 1.1647        | 1.2075                            |
| Isentropic exponent                | 1.1536       | 1.1536       | 1.1630        | 1.2075                            |
| Gas constant                       | 0.3726       | 0.3726       | 0.3714        | 0.3699 kJ/(kg·K)                  |
| Molecular weight                   | 22.3169      | 22.3171      | 22.3856       | 22.4759                           |
| Density                            | 2.1778       | 2.1676       | 1.3402        | 0.1555 kg/m <sup>3</sup>          |
| Sonic velocity                     | 1029.2794    | 1029.1066    | 994.7073      | 829.5779 m/s                      |
| Mach number                        | 0.0000       | 0.0664       | 1.0000        | 2.5840                            |
| Area ratio                         | 9.0000       | 9.0000       | 1.0000        | 4.0000 A/At                       |
| Mass flux                          | 148.0802     | 148.0802     | 1333.1114     | 333.4289 kg/(m <sup>2</sup> ·s)   |
| Viscosity                          | 0.9009       | 0.9007       | 0.8536        | 0.6279 x10 <sup>-4</sup> kg/(m·s) |
| Conductivity, frozen               | 0.2838       | 0.2837       | 0.2626        | 0.1656 W/(m·K)                    |
| Specific heat (p=const), frozen    | 2.4000       | 2.3999       | 2.3641        | 2.1373 kJ/(kg·K)                  |
| Prandtl number, frozen             | 0.7618       | 0.7619       | 0.7685        | 0.8103                            |
| Conductivity, effective            | 0.4235       | 0.4239       | 0.3455        | 0.1677 W/(m·K)                    |
| Specific heat (p=const), effective | 3.1681       | 3.1597       | 3.0587        | 2.1567 kJ/(kg·K)                  |
| Prandtl number, effective          | 0.6740       | 0.6715       | 0.7557        | 0.8076                            |

**Table 3. Combustion Products**

| Product             | Injector mass fraction | Injector mole fraction | Nozzle inlet mass fraction | Nozzle inlet mole fraction | Nozzle throat mass fraction | Nozzle throat mole fraction | Nozzle exit mass fraction | Nozzle exit mole fraction |
|---------------------|------------------------|------------------------|----------------------------|----------------------------|-----------------------------|-----------------------------|---------------------------|---------------------------|
| CO                  | 0.0033693              | 0.0026845              | 0.0033624                  | 0.0026790                  | 0.0016171                   | 0.0012924                   | 0.0000047                 | 0.0000038                 |
| CO2                 | 0.2335327              | 0.1184224              | 0.2335435                  | 0.1184292                  | 0.2362858                   | 0.1201876                   | 0.2388193                 | 0.1219663                 |
| COOH                | 0.0000002              | 0.0000001              | 0.0000002                  | 0.0000001                  |                             |                             |                           |                           |
| H                   | 0.0000116              | 0.0002576              | 0.0000116                  | 0.0002571                  | 0.0000048                   | 0.0001058                   |                           |                           |
| H2                  | 0.0002585              | 0.0028615              | 0.0002580                  | 0.0028567                  | 0.0001349                   | 0.0014978                   | 0.0000008                 | 0.0000088                 |
| H2O                 | 0.6211023              | 0.7694044              | 0.6211151                  | 0.7694291                  | 0.6245109                   | 0.7760102                   | 0.6295429                 | 0.7854187                 |
| H2O2                | 0.0000104              | 0.0000068              | 0.0000103                  | 0.0000068                  | 0.0000052                   | 0.0000034                   |                           |                           |
| HO2                 | 0.0000619              | 0.0000419              | 0.0000617                  | 0.0000417                  | 0.0000329                   | 0.0000223                   | 0.0000006                 | 0.0000004                 |
| O                   | 0.0005855              | 0.0008166              | 0.0005844                  | 0.0008152                  | 0.0002997                   | 0.0004193                   | 0.0000016                 | 0.0000022                 |
| O2                  | 0.1294860              | 0.0903071              | 0.1294874                  | 0.0903090                  | 0.1297413                   | 0.0907638                   | 0.1313989                 | 0.0922943                 |
| OH                  | 0.0115815              | 0.0151971              | 0.0115652                  | 0.0151759                  | 0.0073674                   | 0.0096972                   | 0.0002310                 | 0.0003053                 |
| Gaseous fraction:   | 1.0000000              | 1.0000000              | 1.0000000                  | 1.0000000                  | 1.0000000                   | 1.0000000                   | 1.0000000                 | 1.0000000                 |
| Condensed fraction: | 0.0000000              | 0.0000000              | 0.0000000                  | 0.0000000                  | 0.0000000                   | 0.0000000                   | 0.0000000                 | 0.0000000                 |

**Table 4. Ideal Performance**

| Parameter                    | Sea level | Optimum expansion | Vacuum Units     |
|------------------------------|-----------|-------------------|------------------|
| Characteristic velocity      |           | 1496.4596         | m/s              |
| Effective exhaust velocity   | 2105.6136 | 2143.6214         | 2409.5015 m/s    |
| Specific impulse (by mass)   | 2105.6136 | 2143.6214         | 2409.5015 N·s/kg |
| Specific impulse (by weight) | 214.7128  | 218.5886          | 245.7008 s       |
| Thrust coefficient           | 1.4071    | 1.4325            | 1.6101           |
| Thrust                       | 0.5119    | 0.5211            | 0.5857 kN        |
| Altitude                     | 0.0000    | 1.1128            | km               |
| Ambient pressure             | 1.0000    | 0.8749            | 0.0000 atm       |

**Table 5. Estimated Delivered Performance**

| Parameter                    | Sea level | Optimum expansion | Vacuum Units     |
|------------------------------|-----------|-------------------|------------------|
| Characteristic velocity      |           | 1450.1917         | m/s              |
| Effective exhaust velocity   | 1988.3397 | 2025.1724         | 2282.8320 m/s    |
| Specific impulse (by mass)   | 1988.3397 | 2025.1724         | 2282.8320 N·s/kg |
| Specific impulse (by weight) | 202.7542  | 206.5101          | 232.7841 s       |
| Thrust coefficient           | 1.3711    | 1.3965            | 1.5742           |
| Thrust                       | 0.4834    | 0.4923            | 0.5549 kN        |
| Altitude                     | 0.0000    | 1.1128            | km               |
| Ambient pressure             | 1.0000    | 0.8749            | 0.0000 atm       |

**Table 6. Altitude Performance**

| Altitude, km | Pressure, atm | Specific impulse, N·s/kg | Specific impulse, s | Thrust coefficient | Thrust, kN |
|--------------|---------------|--------------------------|---------------------|--------------------|------------|
| 0.0000       | 1.0000        | 1988.3397                | 202.7542            | 1.3711             | 0.4834     |

**Table 7. Throttled Performance (delivered)**

| Throttle value | Mass flow rate, kg/s | Pressure, MPa | Specific impulse, N·s/kg Sea level | Specific impulse, s Sea level | Thrust, kN Sea level | Specific impulse, N·s/kg Optimum expansion | Specific impulse, s Optimum expansion | Thrust, kN Optimum expansion | Specific impulse, N·s/kg Vacuum | Specific impulse, s Vacuum | Thrust, kN Vacuum |
|----------------|----------------------|---------------|------------------------------------|-------------------------------|----------------------|--------------------------------------------|---------------------------------------|------------------------------|---------------------------------|----------------------------|-------------------|
| 0.2000         | 0.0486               | 0.3988        | 805.8490                           | 82.1737                       | 0.0392               | 2019.0276                                  | 205.8835                              | 0.0982                       | 2278.4444                       | 232.3367                   | 0.1108            |
| 0.2800         | 0.0681               | 0.5588        | 1227.6742                          | 125.1879                      | 0.0836               | 2020.5667                                  | 206.0405                              | 0.1375                       | 2279.5435                       | 232.4487                   | 0.1552            |
| 0.3600         | 0.0875               | 0.7188        | 1462.1705                          | 149.0999                      | 0.1280               | 2021.6262                                  | 206.1485                              | 0.1769                       | 2280.3000                       | 232.5259                   | 0.1996            |
| 0.4400         | 0.1070               | 0.8788        | 1611.4821                          | 164.3254                      | 0.1724               | 2022.4200                                  | 206.2294                              | 0.2163                       | 2280.8667                       | 232.5837                   | 0.2440            |
| 0.5200         | 0.1264               | 1.0389        | 1714.9625                          | 174.8775                      | 0.2168               | 2022.9843                                  | 206.2870                              | 0.2557                       | 2281.2704                       | 232.6248                   | 0.2884            |
| 0.6000         | 0.1459               | 1.1991        | 1790.8332                          | 182.6142                      | 0.2612               | 2023.4993                                  | 206.3395                              | 0.2951                       | 2281.6379                       | 232.6623                   | 0.3328            |
| 0.6800         | 0.1653               | 1.3592        | 1848.8807                          | 188.5334                      | 0.3056               | 2023.9326                                  | 206.3837                              | 0.3346                       | 2281.9471                       | 232.6938                   | 0.3772            |
| 0.7600         | 0.1848               | 1.5194        | 1894.7294                          | 193.2086                      | 0.3501               | 2024.3046                                  | 206.4216                              | 0.3740                       | 2282.2126                       | 232.7209                   | 0.4216            |
| 0.8400         | 0.2042               | 1.6796        | 1931.8621                          | 196.9951                      | 0.3945               | 2024.6292                                  | 206.4547                              | 0.4134                       | 2282.4443                       | 232.7445                   | 0.4661            |
| 0.9200         | 0.2236               | 1.8398        | 1962.5504                          | 200.1244                      | 0.4389               | 2024.9161                                  | 206.4840                              | 0.4529                       | 2282.6490                       | 232.7654                   | 0.5105            |
| 1.0000         | 0.2431               | 2.0000        | 1988.3397                          | 202.7542                      | 0.4834               | 2025.1724                                  | 206.5101                              | 0.4923                       | 2282.8320                       | 232.7841                   | 0.5549            |

**Table 8. Chamber Size**

| Combustion chamber size |         |     |         | Nozzle size      |     |  |  |
|-------------------------|---------|-----|---------|------------------|-----|--|--|
| Dc                      | 45.00   | mm  | Type    | Parabolic nozzle |     |  |  |
| Dt                      | 15.00   | mm  | Rn      | 2.86             | mm  |  |  |
| Lcyl                    | 137.62  | mm  | Tn      | 15.00            | deg |  |  |
| Lc                      | 180.10  | mm  | Te      | 13.50            | deg |  |  |
| L*                      | 1464.07 | mm  | De      | 30.00            | mm  |  |  |
| R1                      | 11.25   | mm  | Le      | 28.64            | mm  |  |  |
| R2                      | 50.36   | mm  | Le/Dt   | 1.91             |     |  |  |
| b                       | 30.00   | deg | Le/Lc15 | 100.97           | %   |  |  |
| Ac/At                   | 9.00    |     | Ae/At   | 4.00             |     |  |  |

  

| Parameter        | Engine    |           | Chamber   |        |
|------------------|-----------|-----------|-----------|--------|
| Thrust           | sea level | 0.4834    | 0.4834    | kN     |
|                  | opt exp   | 0.4923    | 0.4923    | kN     |
|                  | vacuum    | 0.5549    | 0.5549    | kN     |
| Specific Impulse | sea level | 1988.3397 | 1988.3397 | N·s/kg |
|                  | opt exp   | 2025.1724 | 2025.1724 | N·s/kg |
|                  | vacuun    | 2282.8320 | 2282.8320 | N·s/kg |

|                           |              |        |        |      |
|---------------------------|--------------|--------|--------|------|
| Mass flow rate            | total        | 0.2431 | 0.2431 | kg/s |
|                           | oxidizer     | 0.2026 | 0.2026 | kg/s |
|                           | fuel         | 0.0405 | 0.0405 | kg/s |
|                           | film coolant | n.a.   | n.a.   | kg/s |
| <b>Number of chambers</b> |              | 1      |        |      |

**Table 10. Thermal Analysis**